

CASE STUDIES

Pranitha

Name: Mallavarapu Bala

Roll.No:2520030051

Section:1[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files.

```
Tanmayi@Admin:~$ mkdir linux_task1
mkdir: linux_task1: File exists
Tanmayi@Admin:~$ cd linux_task1
Tanmayi@Admin:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
Tanmayi@Admin:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
Tanmayi@Admin:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file1.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file2.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file3.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file4.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug  8 13:38 file5.txt
Tanmayi@Admin:~/linux_task1$ echo "Linux is an operating system." > file1.txt
Tanmayi@Admin:~/linux_task1$ echo "I am learning Linux commands." >> file1.txt
Tanmayi@Admin:~/linux_task1$ cat file1.txt
Linux is an operating system.
I am learning Linux commands.
Tanmayi@Admin:~/linux_task1$ rm file5.txt
Tanmayi@Admin:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
Tanmayi@Admin:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories.

```

Tanmayi@Admin:~$ mkdir linux_task2
mkdir: linux_task2: File exists
Tanmayi@Admin:~$ cd linux_task2
Tanmayi@Admin:~/linux_task2$ touch source.txt data.txt notes.txt
Tanmayi@Admin:~/linux_task2$ nano source.txt
Tanmayi@Admin:~/linux_task2$ mkdir backup
Tanmayi@Admin:~/linux_task2$ ls backup
Tanmayi@Admin:~/linux_task2$ cp source.txt backup/
Tanmayi@Admin:~/linux_task2$ ls backup
source.txt
Tanmayi@Admin:~/linux_task2$ cp data.txt backup/data_backup.txt
Tanmayi@Admin:~/linux_task2$ ls backup
data_backup.txt  source.txt
Tanmayi@Admin:~/linux_task2$ mv notes.txt backup/
Tanmayi@Admin:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
Tanmayi@Admin:~/linux_task2$ mv source.txt original.txt
Tanmayi@Admin:~/linux_task2$ ls
backup  data.txt  original.txt
Tanmayi@Admin:~/linux_task2$ ls -R
.:
backup  data.txt  original.txt

./backup:
data_backup.txt  notes.txt  source.txt
Tanmayi@Admin:~/linux_task2$ tree
Command 'tree' not found, but can be installed with:
sudo apt install tree
Tanmayi@Admin:~/linux_task2$ sudo apt install tree
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 92
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 1s (62.8 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42277 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Tanmayi@Admin:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file

```
Tanmayi@Admin:~$ mkdir linux_task3
Tanmayi@Admin:~$ cd linux_task3
Tanmayi@Admin:~/linux_task3$ touch students.txt
Tanmayi@Admin:~/linux_task3$ ^C
Tanmayi@Admin:~/linux_task3$ ls
students.txt
Tanmayi@Admin:~/linux_task3$ nano students.txt
Tanmayi@Admin:~/linux_task3$ cat students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
Kavya
Rahul
Sneha
Varun
Priya
Rohan
Meera
Aditya
Pooja
Kiran
Tanmayi@Admin:~/linux_task3$ head -n 5 students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
Tanmayi@Admin:~/linux_task3$ head -n 10 students.txt
Aarav
Ananya
Arjun
Diya
Ishaan
Kavya
Rahul
Sneha
Varun
Priya
Tanmayi@Admin:~/linux_task3$ tail -n 5 students.txt
Rohan
Meera
Aditya
Pooja
Kiran
Tanmayi@Admin:~/linux_task3$ tail -n 10 students.txt
Kavya
Rahul
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
Tanmayi@Admin:~$ mkdir organization
Tanmayi@Admin:~$ cd organization
Tanmayi@Admin:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
Tanmayi@Admin:~/organization$ ls
file1.txt file2.txt file3.txt image1.jpg image2.jpg
Tanmayi@Admin:~/organization$ mkdir documents
Tanmayi@Admin:~/organization$ mkdir images
Tanmayi@Admin:~/organization$ ls
documents file1.txt file2.txt file3.txt image1.jpg image2.jpg images
Tanmayi@Admin:~/organization$ mv *.txt documents/
Tanmayi@Admin:~/organization$ ls documents
file1.txt file2.txt file3.txt
Tanmayi@Admin:~/organization$ mv *.jpg images/
Tanmayi@Admin:~/organization$ ls images
image1.jpg image2.jpg
Tanmayi@Admin:~/organization$ ls documents
file1.txt file2.txt file3.txt
Tanmayi@Admin:~/organization$ ls images
image1.jpg image2.jpg
Tanmayi@Admin:~/organization$ mv documents/file1.txt documents/notes.txt
Tanmayi@Admin:~/organization$ ls documents
file2.txt file3.txt notes.txt
Tanmayi@Admin:~/organization$ mv images/image1.jpg images/photo.jpg
Tanmayi@Admin:~/organization$ ls images
image2.jpg photo.jpg
Tanmayi@Admin:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
Tanmayi@Admin:~/organization$ |
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```

Tanmayi@Admin:~$ mkdir project
Tanmayi@Admin:~$ cd project
Tanmayi@Admin:~/project$ mkdir -p source/c source/python docs backup
Tanmayi@Admin:~/project$ ls
backup docs source
Tanmayi@Admin:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:

./source/python:
Tanmayi@Admin:~/project$ touch source/c/file1.c source/c/file2.c source/c/file3.c
Tanmayi@Admin:~/project$ ls source/c
file1.c file2.c file3.c
Tanmayi@Admin:~/project$ touch source/python/file1.py source/python/file2.py
Tanmayi@Admin:~/project$ ls source/python
file1.py file2.py
Tanmayi@Admin:~/project$ ls -R
.:
backup docs source

./backup:

./docs:

./source:
c python

./source/c:
file1.c file2.c file3.c

./source/python:
file1.py file2.py

```

```

./source/python:
file1.py file2.py
Tanmayi@Admin:~/project$ rmdir backup
Tanmayi@Admin:~/project$ ls
docs source
Tanmayi@Admin:~/project$ mkdir backup
Tanmayi@Admin:~/project$ touch backup/test.txt
Tanmayi@Admin:~/project$ ls backup
test.txt
Tanmayi@Admin:~/project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
Tanmayi@Admin:~/project$ rm backup/test.txt
Tanmayi@Admin:~/project$ ls backup
Tanmayi@Admin:~/project$ rmdir backup
Tanmayi@Admin:~/project$ ls
docs source
Tanmayi@Admin:~/project$ tree
.
├── docs
└── source
    ├── c
    │   ├── file1.c
    │   ├── file2.c
    │   └── file3.c
    └── python
        ├── file1.py
        └── file2.py

5 directories, 5 files
Tanmayi@Admin:~/project$ |

```

Task 6: File Permissions Using chmod

Objective: Understand Linux file permissions.

```
Tanmayi@Admin:~$ cd linux_task4
Tanmayi@Admin:~/linux_task4$ touch public.txt private.txt script.sh
Tanmayi@Admin:~/linux_task4$ ls
private.txt  public.txt  script.sh
Tanmayi@Admin:~/linux_task4$ echo "This is a public file." > public.txt
Tanmayi@Admin:~/linux_task4$ echo "This is a private file." > private.txt
Tanmayi@Admin:~/linux_task4$ echo "echo Hello from script" > script.sh
Tanmayi@Admin:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 script.sh
Tanmayi@Admin:~/linux_task4$ chmod 600 private.txt
Tanmayi@Admin:~/linux_task4$ ls -l private.txt
-rw----- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
Tanmayi@Admin:~/linux_task4$ chmod 644 public.txt
Tanmayi@Admin:~/linux_task4$ ls -l public.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
Tanmayi@Admin:~/linux_task4$ chmod +x script.sh
Tanmayi@Admin:~/linux_task4$ ls -l script.sh
-rwxr-xr-x 1 pranitha pranitha 23 Aug 10 06:10 script.sh
Tanmayi@Admin:~/linux_task4$ ls -l
total 12
-rw----- 1 pranitha pranitha 24 Aug 10 06:10 private.txt
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 public.txt
-rwxr-xr-x 1 pranitha pranitha 23 Aug 10 06:10 script.sh
Tanmayi@Admin:~/linux_task4$ chmod -x script.sh
Tanmayi@Admin:~/linux_task4$ ls -l script.sh
-rw-r--r-- 1 pranitha pranitha 23 Aug 10 06:10 script.sh
Tanmayi@Admin:~/linux_task4$ |
```

Task 7: File Permission Challenge

Objective: Understand how permissions affect files and directories.

```
Tanmayi@Admin:~/permission_lab$ ls -l
total 0
-rw-r--r-- 1 pranitha pranitha 0 Aug 10 06:18 test.txt
Tanmayi@Admin:~/permission_lab$ echo "This is a permission test file." > test.txt
Tanmayi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
Tanmayi@Admin:~/permission_lab$ chmod u-r test.txt
Tanmayi@Admin:~/permission_lab$ ls -l test.txt
--w-r--r-- 1 pranitha pranitha 32 Aug 10 06:19 test.txt
Tanmayi@Admin:~/permission_lab$ cat test.txt
cat: test.txt: Permission denied
Tanmayi@Admin:~/permission_lab$ chmod u+r test.txt
Tanmayi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
Tanmayi@Admin:~/permission_lab$ chmod u-w test.txt
Tanmayi@Admin:~/permission_lab$ ls -l test.txt
-r--r--r-- 1 pranitha pranitha 32 Aug 10 06:19 test.txt
Tanmayi@Admin:~/permission_lab$ echo "New line" >> test.txt
-bash: test.txt: Permission denied
Tanmayi@Admin:~/permission_lab$ chmod u+w test.txt
Tanmayi@Admin:~/permission_lab$ echo "New line" >> test.txt
Tanmayi@Admin:~/permission_lab$ cat test.txt
This is a permission test file.
New line
Tanmayi@Admin:~/permission_lab$ chmod u-x test.txt
Tanmayi@Admin:~/permission_lab$ ls -l test.txt
-rw-r--r-- 1 pranitha pranitha 41 Aug 10 06:20 test.txt
Tanmayi@Admin:~/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
Tanmayi@Admin:~/permission_lab$ |
```

Task 8: Backup and Restore

Objective: Create backups and restore files using cp


```

important.txt  linux_task2  linux_task4  organization  permission_lab
Tanmayi@Admin:~$ mv important.txt linux_task6/
Tanmayi@Admin:~$ cd linux_task6
Tanmayi@Admin:~/linux_task6$ ls
important.txt
Tanmayi@Admin:~/linux_task6$ nano important.txt
Tanmayi@Admin:~/linux_task6$ mkdir backup
Tanmayi@Admin:~/linux_task6$ ls
backup  important.txt
Tanmayi@Admin:~/linux_task6$ cp important.txt backup/
Tanmayi@Admin:~/linux_task6$ ls backup
important.txt
Tanmayi@Admin:~/linux_task6$ mv backup/important.txt backup/important_backup.txt
Tanmayi@Admin:~/linux_task6$ ls backup
important_backup.txt
Tanmayi@Admin:~/linux_task6$ rm important.txt
Tanmayi@Admin:~/linux_task6$ ls
backup
Tanmayi@Admin:~/linux_task6$ cp backup/important_backup.txt .
Tanmayi@Admin:~/linux_task6$ ls
backup  important_backup.txt
Tanmayi@Admin:~/linux_task6$ mv important_backup.txt important_restored.txt
Tanmayi@Admin:~/linux_task6$ ls
backup  important_restored.txt
Tanmayi@Admin:~/linux_task6$ cat important_restored.txt
This is important information.
Linux is an operating system.
Files can be backed up.
Backup protects important data.
The cp command copies files.
The mv command moves files.
The rm command deletes files.
Restoring data is useful.
Linux file management is easy.
This is my backup practice.
Tanmayi@Admin:~/linux_task6$ |

```

Task 9: Complete Linux File Management Challenge

Objective: Use almost all the commands together.

```

Tanmayi@Admin:~$ mkdir linux_project
Tanmayi@Admin:~$ cd linux_project
Tanmayi@Admin:~/linux_project$ pwd
/home/pranitha/linux_project
Tanmayi@Admin:~/linux_project$ mkdir source backup docs
Tanmayi@Admin:~/linux_project$ ls
backup docs source
Tanmayi@Admin:~/linux_project$ touch source/main.c source/helper.c source/README.txt
Tanmayi@Admin:~/linux_project$ ls source
README.txt helper.c main.c
Tanmayi@Admin:~/linux_project$ nano source/README.txt
Tanmayi@Admin:~/linux_project$ head -n 5 source/README.txt
Linux is an operating system.
Linux supports file management.
Directories organize files.
The cp command copies files.
The mv command moves files.
Tanmayi@Admin:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
Tanmayi@Admin:~/linux_project$ tail -n 5 source/README.txt
C is a programming language.
Source files contain program code.
Backup protects important files.
Documentation explains a project.
This is my Linux project.
Tanmayi@Admin:~/linux_project$ cp source/main.c backup/
Tanmayi@Admin:~/linux_project$ ls backup
main.c
Tanmayi@Admin:~/linux_project$ cp source/README.txt docs/
Tanmayi@Admin:~/linux_project$ ls docs
README.txt

```

```

Tanmayi@Admin:~/linux_project$ ls docs
README.txt
Tanmayi@Admin:~/linux_project$ mv source/helper.c source/functions.c
Tanmayi@Admin:~/linux_project$ ls source
README.txt functions.c main.c
Tanmayi@Admin:~/linux_project$ chmod 644 source/README.txt
Tanmayi@Admin:~/linux_project$ ls -l source/README.txt
-rw-r--r-- 1 pranitha pranitha 672 Aug 10 06:53 source/README.txt
Tanmayi@Admin:~/linux_project$ rm backup/main.c
Tanmayi@Admin:~/linux_project$ ls backup
Tanmayi@Admin:~/linux_project$ rmdir backup
Tanmayi@Admin:~/linux_project$ ls
docs source
Tanmayi@Admin:~/linux_project$ ls -R
.:
docs source

./docs:
README.txt

./source:
README.txt functions.c main.c
Tanmayi@Admin:~/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
Tanmayi@Admin:~/linux_project$ |

```

Task 10: Mini Linux Administration Challenge

Objective: Complete a real-world-style file management exercise.

```
Tanmayi@Admin:~$ mkdir student_portal
Tanmayi@Admin:~$ cd student_portal
Tanmayi@Admin:~/student_portal$ pwd
/home/pranitha/student_portal
Tanmayi@Admin:~/student_portal$ mkdir src docs backup logs config
Tanmayi@Admin:~/student_portal$ ls
backup config docs logs src
Tanmayi@Admin:~/student_portal$ touch src/main.c src/student.c docs/README.txt config/config.txt logs/system.log
Tanmayi@Admin:~/student_portal$ ls -R
.:
backup config docs logs src

./backup:

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c student.c
Tanmayi@Admin:~/student_portal$ nano logs/system.log
Tanmayi@Admin:~/student_portal$ head -n 5 logs/system.log
System started successfully.
Student portal initialized.
Database connection established.
User login service started.
Student records loaded.
Tanmayi@Admin:~/student_portal$ tail -n 5 logs/system.log
Profile updated successfully.
Student searched successfully.
Report generated.
Backup process completed.
System running normally.
Tanmayi@Admin:~/student_portal$ cp docs/README.txt backup/
Tanmayi@Admin:~/student_portal$ ls backup
README.txt
```

```
Tanmayi@Admin:~/student_portal$ ls backup
README_backup.txt
Tanmayi@Admin:~/student_portal$ chmod 600 config/config.txt
Tanmayi@Admin:~/student_portal$ ls -l config/config.txt
-rw----- 1 pranitha pranitha 0 Aug 10 07:00 config/config.txt
Tanmayi@Admin:~/student_portal$ chmod 644 docs/README.txt
Tanmayi@Admin:~/student_portal$ ls -l docs/README.txt
-rw-r--r-- 1 pranitha pranitha 0 Aug 10 07:00 docs/README.txt
Tanmayi@Admin:~/student_portal$ rm backup/README_backup.txt
Tanmayi@Admin:~/student_portal$ ls backup
Tanmayi@Admin:~/student_portal$ rmdir backup
Tanmayi@Admin:~/student_portal$ ls
config docs logs src
Tanmayi@Admin:~/student_portal$ ls -R
.:
config docs logs src

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
main.c student.c
Tanmayi@Admin:~/student_portal$ tree
.
├── config
│   └── config.txt
├── docs
│   └── README.txt
├── logs
│   └── system.log
└── src
    ├── main.c
    └── student.c

5 directories, 5 files
```